

Computer Architecture

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Computer architecture: Computer architecture is the theory behind the operational design of a computer system.

Computer architecture is the science and art of selecting and interconnecting hardware components to create computers that meet functional, performance & cost goals.

⇒ computer architecture is a description of the structure of a computer system that includes the instruction set architecture design, logic design and implementation of the instruction.

Subcategories of CA:

- ① Instruction set architecture: It defines the machine code that a processor reads and acts upon as well as the word size, memory address modes, processor registers & data type.

② Microarchitecture: This is also known as computer organization that describes how a particular processor will implement the ISA.

③ System design: Includes all of the other hardware components within a computing system, such as data processing other than the CPU, virtualization and multiprocessing.

Types of Architecture:

- ① RISC (Reduced Instruction set computer)
- ② CISC (Complex instruction set computer)
- ③ Harvard Architecture
- ④ Modified harvard architecture
- ⑤ Superscalar architecture
- ⑥ Vector processing architecture
- ⑦ Multithreading Architecture
- ⑧ Von Neumann Architecture

Importance of Computer Architecture:

- ⇒ To understand the relationship between HLL and machine language.
- ⇒ Getting more performance by performing operations in parallel.
- ⇒ Using abstraction to simplify design.
- ⇒ The memory hierarchy model helps to optimize the page file

Cache / RAM etc. to execute in least time.
→ To define hardware and software interface a system.

Computer architecture Vs Organization:

Architecture	Organization
① It describes what computer does.	① It describes how computer does a task.
② It deals with high level design & issues.	② It deals with low level design issues.
③ Interface between H/W & S/W	③ Deals with components of connection in a system.
④ while designing computer system architecture is considered 1st.	④ Organization is done on the basis of architecture.
⑤ Abstract model and is programmer's view in terms of instructions, addressing modes & registers.	⑤ It expresses the realization of architecture.

Difference between RISC and CISC:

RISC	CISC
① Emphasis on software	① Emphasis on hardware
② Small number of fixed length instructions,	② Large number of instructions
③ Simple standardised instructions	③ Complex variable length instructions
④ single clock cycle instructions	④ Instructions can have several clock cycle.
⑤ Heavy use of RAM	⑤ More efficient use of RAM
⑥ Low cycles per second with large code size	⑥ small code sizes with high cycles per second.

between